

Programme: Diploma in Cloud Computing and BigData	
Course Code: 6279B	Course Title: Artificial Neural Network Lab
Semester: 6	Credits: 1.5
Course Category: Programme elective course	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- Exposure to the Artificial Neural Network (ANN) techniques

Course Prerequisites:

Topic	Course Code	Course name	Semester
Problem Analysis and Programming concept	2131	Problem Solving and Programming	2
Programming in Python	3279	Programming in Python Lab	3
Implementation of data structures	4278	Data Structures and Algorithms Lab	4
Concept of Machine Learning Techniques		Machine Learning	4
Implementation of Machine Learning Algorithms for Prediction and Classification problems		Machine Learning Lab	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Configure Development Environment setup	7	Applying
CO2	Practice Data Preprocessing techniques	12	Applying

CO3	Demonstrate Convolution Neural Network	12	Applying
CO4	Experiment with pre trained Convolutional Neural Network models	8	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2		3			
CO2	3	2		3			
CO3	3	2		3			
CO4	3	2		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Configure Development Environment setup		
M1.01	Practice installation and configuration of Development Environment with required Libraries	2	Applying
M1.02	Experiment basic operations for Machine Learning with Python libraries.	5	Applying
Contents: Google Colab- Python Environments- Anaconda/Miniconda/Jupyter Notebook/Pytorch - Installation and configuration of IDE and required libraries. Python Libraries for Machine Learning- Numpy, Scikit Learn, Pandas, Matplotlib, Tensorflow, keras, NLTK, CV2, PIL			
CO2	Practice Data Preprocessing techniques		
M2.01	Practice data pre-processing on text data	6	Applying

M2.02	Perform Image preprocessing	6	Application
	Lab Test – I	3	
Contents : Read csv file- missing data handling, null data handling, removing irrelevant data, avoid duplication, scaling, standardization, normalization, dimensionality reduction Read image files- image preprocessing- resize, scale, crop, color change			
CO3	Demonstrate Convolution Neural Network		
M3.01	Practice Deep Neural Network for Classification using feature set (text data)	6	Applying
M3.02	Experiment Convolutional Neural Network for Classification using image data	6	Applying
Contents: Split data for training and testing, Feature scaling, standardization, Performance evaluation Example for problems- Sentiment analysis, Population prediction, Face detection, Hand written digit prediction, Dog-cat classification.			
CO4	Experiment with pre trained Convolutional Neural Network models		
M4.01	Practice with pre trained CNN models- VGG, GoogleNet, ResNet and Xception	4	Applying
M4.02	Experiment pretrained models -VGG, GoogleNet, ResNet and Xception	4	Applying
	Series Test – II	3	
Contents : Confusion matrix, Accuracy, Recall, F1-Score			

Text / Reference

T/R	Book Title/Author
R1	Deep Learning by Ian Goodfellow, Yoshua Bengio and Aaron Courville, The MIT Press
R2	Machine Learning by Tom Mitchell, McGraw-Hill Press, 1997